Machine Learning Final Project Summary

Team Formation

- Form your own team
- Each group should have 1–4 students

Topic Selection

- Choose your own machine learning related topic
- Only machine learning algorithms taught in this course are allowed

Deep Learning methods are NOT allowed
(such as CNN, RNN, Transformer)

Model Requirements

At least 3 different models must be compared

The project must include:

- Accuracy comparison
- Confusion Matrix analysis
- Feature Importance analysis
- Comparison of model advantages and disadvantages

Week 16 – Project Oral Report

Each group will give:

8–10 minutes presentation

Submission Items

1. PowerPoint Slides (PPT)
2. Presentation Explanation Video
3. Dataset (raw data + processed data)
4. Model (training results)

Grading Policy

Data Collection and Preprocessing: 35%

Model Construction and Result Analysis: 35%

Overall Completeness: 30%

Bonus Items

- Self-collected images/data
- Web scraping by yourself
- Creative project topic
- System demonstration
- Data visualization

Most Important Reminders

No plagiarism

Do NOT simply download a dataset and run it directly

Your own analysis is required

Full Project Description Below

Machine Learning Final Project

Project Description

The purpose of this final project is to allow students to apply the machine learning methods learned in this course, including data collection, data preprocessing, model construction, model training, result analysis, and final presentation.

This project is designed to help students experience the complete workflow of a machine learning project.

The standard machine learning process introduced in class is:

Project Problem Analysis → Data Preprocessing → Feature Engineering → Selection Algorithm → Model Training → Model Evaluation

Please complete your final project based on this workflow.

I. Project Requirements

1. Team Formation and Topic Selection

Please form your own team and discuss your project topic.

Suggested topics include:

- Student grade prediction
- House price prediction
- Handwritten digit recognition
- Customer purchasing behavior analysis
- Medical disease classification
- Weather prediction
- Stock trend classification
- Image classification
- Animal species classification
- Iris classification (extended version)
- Go player strength prediction
- Social media sentiment analysis

2. Dataset

Datasets may be obtained using the following methods:

Method 1: Self-Collected Data (Bonus)

For example:

- Taking your own photos
- Questionnaire surveys
- Sensor data collection
- Interview data整理
- Designing your own data recording tables

Method 2: Online Data Collection

For example:

- Kaggle
- UCI Machine Learning Repository
- Government Open Data Platforms
- Web scraping

Method 3: Download Existing Datasets

For example:

- WEKA built-in datasets
- Kaggle
- UCI datasets

※ It is strongly recommended not to directly use a completely unprocessed dataset.

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3. Group Size

Each group should have:

1–4 students

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4. Restrictions

Only machine learning algorithms taught in this course are allowed

Allowed Algorithms

- Decision Tree (J48)
- Random Forest
- Linear Regression
- Logistic Regression
- KNN (IBk)
- Naive Bayes
- Bayesian Network
- SVM
- Ensemble Methods
- Bagging
- Boosting
- LogitBoost
- Clustering (K-means)

Not Allowed

- CNN
- RNN
- LSTM
- Transformer
- Deep Learning Frameworks (PyTorch / TensorFlow)

(Unless special permission is given by the instructor)

II. Project Oral Report

Presentation Time

The final project oral report will be held in:

Week 16

Please prepare your complete presentation and final results in advance.

Presentation Duration

Each group:

8–10 minutes presentation

III. Presentation Requirements

Please prepare a:

PowerPoint Presentation

The presentation must include the following sections:

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1. Project Motivation

Explain:

- Why did you choose this topic?
- Why is this problem important?
- What is its practical value?

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2. Project Objective

Clearly define:

- What problem are you solving?
- What are you predicting?
- What are you classifying?
- What is your final goal?

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3. Dataset Description

Must include:

- Data source
- Number of instances
- Number of attributes
- Class label
- Data format (CSV / ARFF / Image)

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4. Data Preprocessing

For example:

- Missing value handling

- Noise removal
- Normalization
- Feature selection
- Attribute transformation
- Data cleaning
- Label encoding

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5. Model Construction

Explain:

- Which models are used (at least 3 models)
- Why these models were selected
- Model architecture
- Important parameter settings

For example:

- J48: confidenceFactor, minNumObj
- Random Forest: numTrees, maxDepth
- IBk: k value
- Logistic: ridge
- LogitBoost: base learner

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6. Performance Evaluation

For Classification

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve
- AUC

For Regression

- MAE
- RMSE
- Correlation coefficient

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7. Model Comparison (Important)

At least 3 models must be compared

Example:

Model	Accuracy
J48	85%
Random Forest	91%
LogitBoost	93%

Please analyze:

- Which model performs best?
- Why?
- Is there any overfitting?
- What is the difference in model complexity?

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8. Final Result / Demo

Please demonstrate:

- Prediction results
- Classification results
- System screenshots
- WEKA execution results
- Data visualization charts

Text-only explanation is NOT sufficient.

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9. Division of Work

Clearly explain the work responsibilities of each team member.

IV. Submission

Please submit the following items:

(1) PowerPoint Slides (PPT)

(2) Presentation Explanation Video

Recommended length:

5–10 minutes

(3) Dataset

Raw data + processed data

(4) Model

For example:

- WEKA model
- Training results

(5) Others (such as Code)

V. Grading Policy

1. Data Collection and Preprocessing (35%)

Self-collected data + preprocessing

15–35 points

Online collected data + preprocessing

10–35 points

Existing dataset + no preprocessing

0–20 points

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2. Model Construction & Result Analysis (35%)

3 or more models + detailed analysis

20–35 points

2 models

10–25 points

Only 1 model

0–20 points

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3. Overall Completeness (30%)

Including:

- Motivation
- Creativity
- Presentation quality
- Analysis depth
- Completeness of final demonstration
- Teamwork and division of labor

VI. Bonus Items

Additional bonus points may be given for:

- Self-collected data
- Web scraping
- Creative project topics
- Visualization analysis
- Complete system demonstration
- Complete model comparison

VII. Important Notes

- No plagiarism
- Do NOT simply download a dataset and run it directly
- Your own analysis and discussion are required
- Final demo is required
- Model comparison is required
- Only course-taught algorithms are allowed
- Final grading will be conducted in Week 16

Please start early and avoid rushing during the final week.

VIII. Example Topics

- House rental price prediction in Hualien
- Student grade prediction
- Mobile phone price classification
- Diabetes prediction
- Insurance purchase prediction
- Customer churn analysis
- Advanced Iris classification
- Stock price trend classification
- Go player strength prediction
- Movie review sentiment analysis